

# SAT UZBEKISTAN

Telegram @satuzbhunters

1



Mark for Review



Which expression is equivalent to  $6xy(2x^2 + 5y)$ ?

(A)  $8x^3y + 11xy^2$

(B)  $12x^2y + 5y$

(C)  $12x^2y + 30xy$

(D)  $12x^3y + 30xy^2$

2



Mark for Review



Which expression is equivalent to  $7x^6 - 2x^5 + 9x^4$ ?

(A)  $x^5(7x - 2)$

(B)  $x^4(7x^2 - 2x + 9)$

(C)  $9x^4(7x^2 - 2x + 1)$

(D)  $2x^6(-2x^5 + 9x^4 + 1)$

3



Mark for Review



The height of a certain vehicle is 61 inches. The vehicle's length is 3 times its height. Which of the following systems of equations represents this situation, where  $x$  is the length of the vehicle, in inches, and  $y$  is the height of the vehicle, in inches?

(A)  $y = 3x$   
 $x = 61$

(B)  $y = 3x$   
 $y = 61$

(C)  $x = 3y$   
 $x = 61$

(D)  $x = 3y$   
 $y = 61$

4



Mark for Review



For the linear function  $f$ , the graph of  $y = f(x)$  in the  $xy$ -plane has a slope of  $-54$  and passes through the point  $(0, 0)$ . Which equation defines  $f$ ?

(A)  $f(x) = -54x$

(B)  $f(x) = -36x$

(C)  $f(x) = -18x$

(D)  $f(x) = x$

5



Mark for Review



The function  $g$  is defined by  $g(x) = \frac{5}{9}x - \frac{7}{9}$ . What is the value of  $g(18)$ ?

(A)  $\frac{38}{9}$

(B)  $\frac{52}{9}$

(C)  $\frac{83}{9}$

(D)  $\frac{97}{9}$

Directions ▾

Hide

## Student-produced response directions

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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

## Examples

| Answer | Acceptable ways to enter answer | Unacceptable; will NOT receive credit |
|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | $31/2$<br>3 1/2                       |

6



Mark for Review

The function  $f$  is defined by  $f(x) = x^2 + x + 61$ . What is the value of  $f(2)$ ?

A) 61   B) 67   C) 69   D) 71

Answer Preview:



7



Mark for Review



$$f(x) = 5x + 7$$

The function  $f$  gives the estimated height, in feet, of a poplar tree  $x$  years after its height was first measured. Which statement is the best interpretation of 7 in this context?

- (A) The tree will be measured each year for 7 years.
- (B) The tree is estimated to grow to a maximum height of 7 feet.
- (C) The estimated height of the tree increased by 7 feet each year.
- (D) The estimated height of the tree was 7 feet when it was first measured.

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8



Mark for Review

In the  $xy$ -plane, line  $k$  has a slope of 5 and a  $y$ -intercept of  $(0, -35)$ . What is the  $x$ -coordinate of the  $x$ -intercept of line  $k$ ?

A) 5    B) -5    C) 7    D) -7

Answer Preview:

- If your answer is a **mixed number** (such as  $3\frac{1}{2}$ ), enter it as an improper fraction ( $7/2$ ) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

## Examples

| Answer         | Acceptable ways to enter answer           | Unacceptable: will NOT receive credit |
|----------------|-------------------------------------------|---------------------------------------|
| 3.5            | 3.5<br>3.50<br>7/2                        | $31/2$<br>3 1/2                       |
| $\frac{2}{3}$  | $2/3$<br>.6666<br>.6667<br>0.666<br>0.667 | 0.66<br>.66<br>0.67<br>.67            |
| $-\frac{1}{3}$ | -1/3<br>-.3333<br>-0.333                  | -.33<br>-0.33                         |

9



Mark for Review

$$\frac{27(k+6)}{3(k+6)}$$

The given expression is equivalent to  $c$ , where  $c$  is a constant and  $k > -6$ . What is the value of  $c$ ?

A) 9    B) -9    C) 27    D) 3

Answer Preview:



10



Mark for Review



The table gives the perimeters of similar triangles  $TUV$  and  $XYZ$ , where  $\overline{TU}$  corresponds to  $\overline{XY}$ . The length of  $\overline{TU}$  is 6.

|                | Perimeter |
|----------------|-----------|
| Triangle $TUV$ | 50        |
| Triangle $XYZ$ | 150       |

What is the length of  $\overline{XY}$ ?

(A) 2

(B) 6

(C) 18

(D) 56

**11**

Mark for Review



If  $9(7 - 8x) + 2 = 8(7 - 8x) + 17$ , what is the value of  $7 - 8x$ ?

(A)  $-15$

(B)  $-1$

(C)  $1$

(D)  $15$



$$x + 33 = y$$

$$(x + 33)^2 = y$$

A solution to the given system of equations is  $(x, y)$ . What is a possible value of  $x$ ?

(A)  $-32$

(B)  $0$

(C)  $1$

(D)  $33$



A circle in the  $xy$ -plane has the equation  $(x - 14)^2 + (y - k)^2 = 36$ . Which of the following gives the center of the circle and its radius?

- (A) The center is at  $(14, k)$  and the radius is 6.
- (B) The center is at  $(k, 14)$  and the radius is 6.
- (C) The center is at  $(k, 14)$  and the radius is 36.
- (D) The center is at  $(14, k)$  and the radius is 36.



14



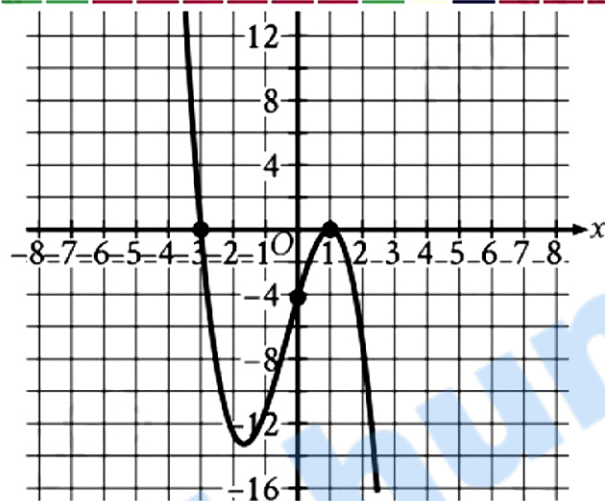
Mark for Review



The town of Fillmore is planning its annual fireworks show. The equation  $y = -16(x - 6.8)^2 + 740$  gives the estimated height  $y$ , in feet, of a type of firework  $x$  seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the  $xy$ -plane?

- (A) This type of firework reaches an estimated maximum height of 740 feet 16 seconds after it is launched into the air.
- (B) This type of firework reaches an estimated maximum height of 740 feet 6.8 seconds after it is launched into the air.
- (C) This type of firework reaches an estimated maximum height of 16 feet 740 seconds after it is launched into the air.
- (D) This type of firework reaches an estimated maximum height of 6.8 feet 740 seconds after it is launched into the air.

Hide



The graph of  $y = f(x)$  is shown where  $f(x) = ax^3 + bx^2 + cx + d$  and  $a$ ,  $b$ ,  $c$ , and  $d$  are constants. For how many distinct values of  $x$  does  $f(x) = 0$ ?

(A) One

(B) Two

(C) Three

(D) Four



In the  $xy$ -plane, circle  $M$  is the graph of the equation  $(x - 3)^2 + (y - 6)^2 = 4$ . Circle  $P$  has the same center as circle  $M$  but has a radius that is twice the radius of circle  $M$ . Which equation represents circle  $P$ ?

(A)  $(x - 3)^2 + (y - 6)^2 = 8$

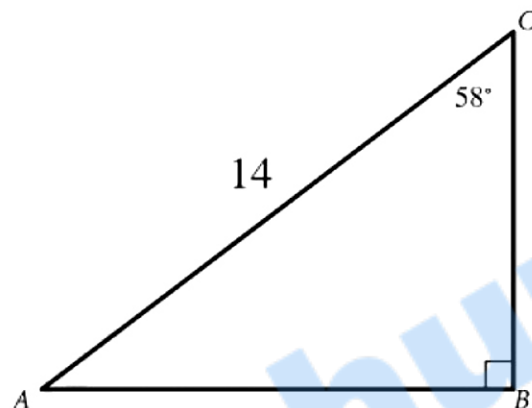
(B)  $(x - 3)^2 + (y - 6)^2 = 16$

(C)  $(x - 6)^2 + (y - 12)^2 = 4$

(D)  $(x - 6)^2 + (y - 12)^2 = 8$



Mark for Review



Note: Figure not drawn to scale.

Which expression represents the length of line segment  $AB$ ?

(A)  $14 \sin 58^\circ$

(B)  $\frac{14}{\sin 58^\circ}$

(C)  $14 \cos 58^\circ$

(D)  $\frac{14}{\cos 58^\circ}$

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|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | $31/2$<br>$3\ 1/2$                    |

18

Mark for Review

The product of a positive number  $x$  and the number that is 15 less than  $x$  is equal to 286. What is the value of  $x$ ?

A) 6    B) 26    C) 24    D) 32

Answer Preview:

## Student-produced response directions

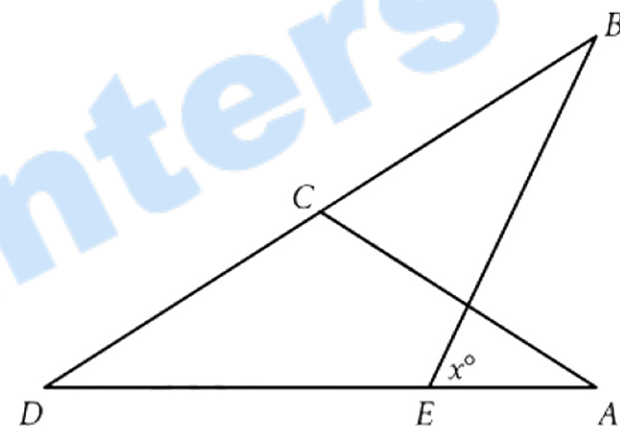
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## Examples

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|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | $3\frac{1}{2}$<br>3 1/2               |

19

Mark for Review



Note: Figure not drawn to scale.

In the figure,  $AC = CD$ . The measure of angle  $EBC$  is  $29^\circ$ , and the measure of angle  $ACD$  is  $108^\circ$ . What is the value of  $x$ ?

A) 74   B) 62   C) 68   D) 65

Answer Preview:

## Student-produced response directions

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### Examples

| Answer | Acceptable ways to enter answer | Unacceptable: will NOT receive credit |
|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | $31/2$<br>$3\ 1/2$                    |

20



Mark for Review

The positive number  $a$  is 2,800% of the number  $c$ , and  $c$  is 25% of the number  $b$ . If  $a - b = wc$ , where  $w$  is a constant, what is the value of  $w$ ?

A) 16    B) 20    C) 24    D) 28

**Answer Preview:**

## Student-produced response directions

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## Examples

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|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | $3\frac{1}{2}$<br>$3\ 1/2$            |

21

Mark for Review

Line  $j$  is defined by  $4x + 5y = 55$ . Line  $k$  is parallel to line  $j$  in the  $xy$ -plane. An equation of line  $k$  is  $24x + ry = 15$ , where  $r$  is a constant. If line  $k$  passes through the point  $(0, b)$ , what is the value of  $b$ ?

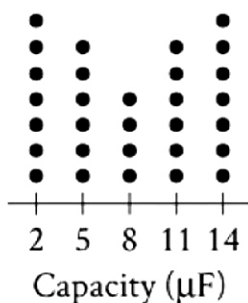
A)  $1/4$    B)  $1/2$    C)  $1/3$    D)  $2/3$ 

Answer Preview:

22

Mark for Review

APC



The dot plot shows the distribution of capacity for a set of capacitors, set A, which a researcher used for a certain experiment. For another experiment, the researcher used a different set of capacitors, set B. Set B has the same number of capacitors as set A but the capacity of each capacitor of set B is 17 microfarads ( $\mu\text{F}$ ) greater than the capacity of each respective capacitor in set A. Which of the following is true about the capacities of the capacitors in set B?

- (A) The mean capacity is 8  $\mu\text{F}$ , and the range of capacities is 12  $\mu\text{F}$ .
- (B) The mean capacity is 8  $\mu\text{F}$ , and the range of capacities is 29  $\mu\text{F}$ .
- (C) The mean capacity is 25  $\mu\text{F}$ , and the range of capacities is 12  $\mu\text{F}$ .
- (D) The mean capacity is 25  $\mu\text{F}$ , and the range of capacities is 29  $\mu\text{F}$ .

# SAT UZBEKISTAN

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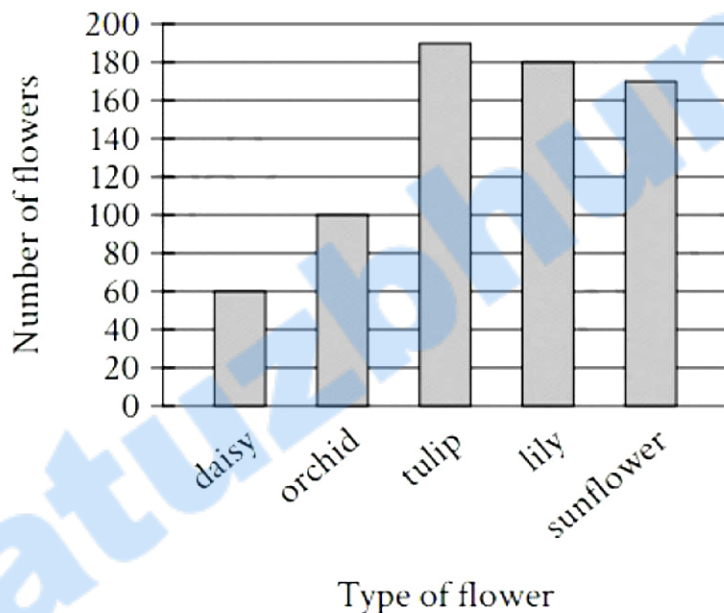
1



Mark for Review

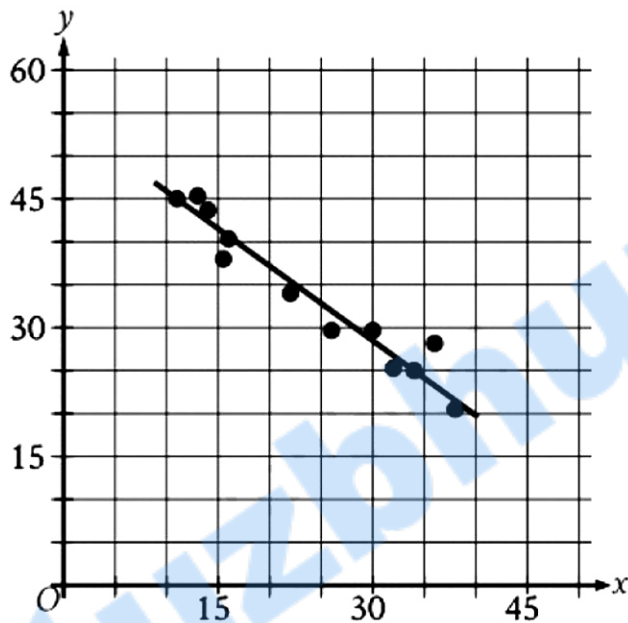


The bar graph shows the distribution of flower type for the 700 flowers that a florist has in stock.



For which two types of flowers is the total number of flowers of the two types exactly  $\frac{2}{5}$  of the florist's stock?

- A) daisy and tulip    B) orchid and lily    C) tulip and lily    D) sunflower and orchid



The scatterplot shows the relationship between two variables,  $x$  and  $y$ , for data set A. A line of best fit for the data is also shown. Data set B is created by subtracting 11 units from the value of  $y$  for each data point from data set A. Which of the following is closest to the  $y$ -coordinate of the  $y$ -intercept of the line of best fit for data set B?

- A) 62.5   B) 43.5   C) 15.5   D) 35.5

3



Mark for Review



If  $4(2x + 11) - 2(2x - 5) = -9 + 13x$ , what is the value of  $12x$ ?

(A)  $\frac{7}{12}$

(B) 7

(C) 12

(D) 84

4



Mark for Review



An object has a mass of 308 grams and a volume of 28 cubic centimeters. What is the density, in grams per cubic centimeter, of the object?

(A) 11

(B) 280

(C) 336

(D) 8,624

5



Mark for Review



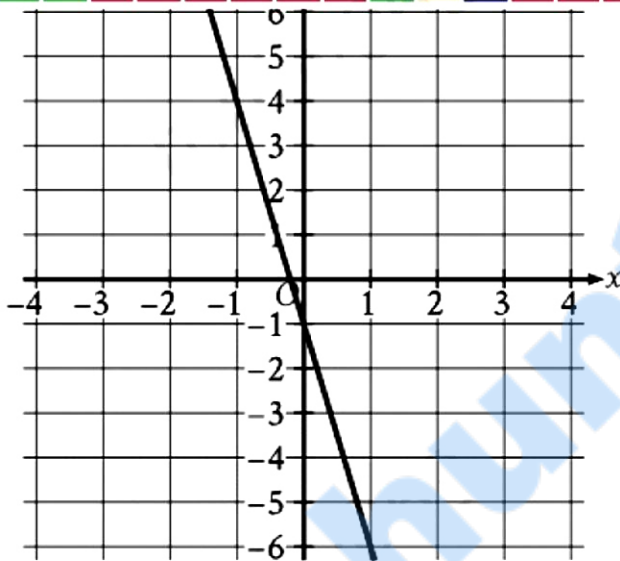
In the  $xy$ -plane, which of the following does NOT contain any points that are part of the solution set to  $3x - 7y > 21$ ?

(A) The region where  $x < 0$  and  $y > 0$

(B) The region where  $x < 0$  and  $y < 0$

(C) The region where  $x > 0$  and  $y > 0$

(D) The  $x$ -axis



Which equation defines the linear function  $f$ ?

(A)  $f(x) = -14x - 9$

(B)  $f(x) = -5x - 9$

(C)  $f(x) = -5x + 8$

(D)  $f(x) = -14x + 8$

7



Mark for Review



In triangle  $FGH$  and triangle  $KLM$ , the measures of angles  $G$  and  $L$  are each  $30^\circ$ . The lengths of  $\overline{GH}$  and  $\overline{LM}$  are each 34 centimeters, and  $\frac{GH}{GF} = \frac{LM}{LK}$ . Which additional piece of information would be necessary to prove that triangle  $FGH$  is congruent to triangle  $KLM$ ?

- (A) The lengths of  $\overline{FG}$  and  $\overline{KM}$  are each 17 centimeters.
- (B) The measures of angles  $H$  and  $M$  are each  $60^\circ$ .
- (C) The measures of angles  $F$  and  $K$  are each  $90^\circ$ .
- (D) No additional information is necessary.

8



Mark for Review



$$y = 3x + 9$$

$$3y = 8x - 6$$

The solution to the given system of equations is  $(x, y)$ . What is the value of  $x - y$ ?

(A)  $-123$

(B)  $-33$

(C)  $3$

(D)  $57$

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9

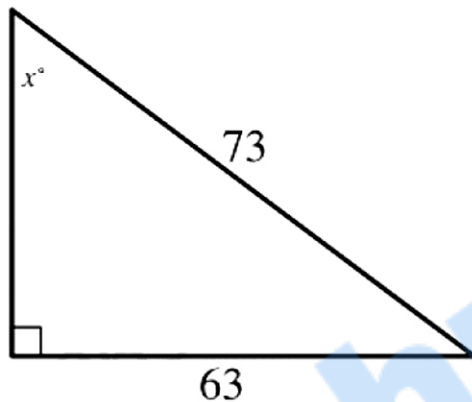


Mark for Review

An exponential function  $f$  is defined by  $f(x) = c^x$ , where  $c$  is a constant greater than 1. If  $f(6) = 9 \cdot f(4)$ , what is the value of  $c$ ?

A) 1   B) 3   C) 4   D) 6

Answer Preview:



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of  $\sin x^\circ$ ?

(A)

$$\frac{1}{73}$$

(B)

$$\frac{10}{73}$$

(C)

$$\frac{63}{73}$$

(D)

$$\frac{136}{73}$$

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11

Mark for Review

$$-x - wy = -337$$

$$2x - wy = 47$$

In the given system of equations,  $w$  is a constant. In the  $xy$ -plane, the graphs of these equations intersect at the point  $(q, 19)$ , where  $q$  is a constant. What is the value of  $w$ ?

A) 13    B) 11    C) 9    D) 7

Answer Preview:



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## Examples

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12



Mark for Review

|              | Site A | Site B | Total |
|--------------|--------|--------|-------|
| Red maple    | 35     | 15     | 50    |
| Chestnut oak | 31     | 20     | 51    |
| Total        | 66     | 35     | 101   |

The table shows the distribution of two types of trees at two different sites. If a tree represented in the table is selected at random, what is the probability of selecting a tree from site A, given that the tree is a red maple? (Express your answer as a decimal or fraction, not as a percent.)

A) 35/101    B) 35/66    C) 35/50    D) 66/101

Answer Preview:

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13

Mark for Review

A beaker containing a liquid is placed on a table. The function  $g(t) = 295 + (361 - 295)(2.72)^{-0.104t}$  gives the approximate temperature, in kelvins, of the liquid  $t$  minutes after the beaker was placed on the table. According to this function, what was the approximate temperature, in kelvins, of the liquid when the beaker was placed on the table?

A) 295    B) 66    C) 361    D) 0

Answer Preview:

14



Mark for Review



For the exponential function  $f$ , the value of  $f(1)$  is  $k$ , where  $k$  is a constant. Which of the following equivalent forms of the function  $f$  shows the value of  $k$  as the coefficient or the base?

(A)  $f(x) = 49(1.7)^{x+1}$

(B)  $f(x) = 85(1.7)^x$

(C)  $f(x) = 144.5(1.7)^{x-1}$

(D)  $f(x) = 245.65(1.7)^{x-2}$

15



Mark for Review



$$y = 9\left(\frac{a}{6}\right)^{x+c} - b$$

How many times does the graph of the given equation in the  $xy$ -plane cross the  $x$ -axis, where  $a$ ,  $b$ , and  $c$  are positive constants such that  $a > 6$  and  $b > c$ ?

(A) Zero

(B) One

(C) Two

(D) Three

16



Mark for Review



For a polynomial function  $f$ , the graph of  $y = f(x)$  in the  $xy$ -plane contains the points  $(-6, 0)$ ,  $(7, 0)$ ,  $(0, 0)$ , and  $(4, 0)$ . Which of the following must be a factor of  $f(x)$ ?

(A)  $x^2 - 2x - 24$

(B)  $x^2 - x + 42$

(C)  $x^2 - 11x - 28$

(D)  $x^2 - 7x$

Directions ▾

Hide

Student-produced response directions

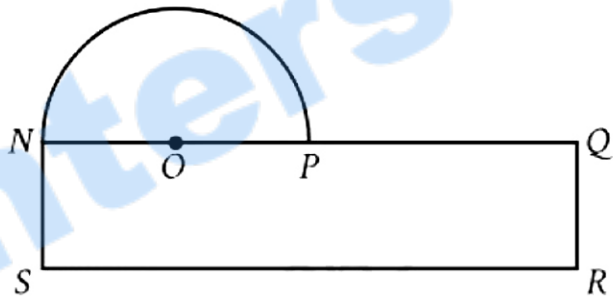
- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as  $3\frac{1}{2}$ ), enter it as an improper fraction ( $7/2$ ) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

| Answer | Acceptable ways to enter answer | Unacceptable: will NOT receive credit |
|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | $31/2$<br>$3\ 1/2$                    |

17

Mark for Review



Note: Figure not drawn to scale.

In the given figure,  $\overline{NP}$  is the diameter of semicircle  $O$ . Angles  $Q$ ,  $R$ , and  $S$  are right angles, and point  $P$  is the midpoint of  $\overline{NQ}$ . If  $PQ = 52$  feet and  $NS = 18$  feet, the area of the figure can be expressed as  $a\pi + b$  square feet, where  $a$  and  $b$  are integers. What is the value of  $a - b$ ?

A) -1534    B) -1542    C) -1536    D) -1544

Answer Preview:

18



Mark for Review



$$\frac{1}{5xy} + xyz = \frac{1}{4yz}$$

In the given equation,  $x$ ,  $y$ , and  $z$  are positive numbers. Which expression is equivalent to  $y$ ?

(A)

$$\frac{5x-4z}{20x^2z^2}$$

(B)

$$\sqrt{\frac{5x-4z}{20x^2z^2}}$$

(C)

$$\frac{1}{4xz^2-5x^2z}$$

(D)

$$\sqrt{\frac{1}{4xz^2-5x^2z}}$$



19



Mark for Review



The equation  $2|x - 5| = k$ , where  $k$  is a constant, has exactly one solution. Which of the following could be the value of  $\frac{k}{2}$ ?

(A)  $-5$  only

(B)  $-5$  or  $5$

(C)  $0$  only

(D)  $5$  only



20



Mark for Review



The percent increase in mass of a certain red kangaroo from 80 days old to 160 days old was 676%. If this red kangaroo's mass was  $k$  grams at 80 days old, which expression represents its mass, in grams, at 160 days old?

(A)  $7.76k$

(B)  $6.76k$

(C)  $1.08k$

(D)  $0.08k$

21



Mark for Review



| $x$ | $g(x)$ |
|-----|--------|
| -21 | 2      |
| -9  | 0      |
| 15  | 4      |

The table shows three values of  $x$  and their corresponding values of  $g(x)$ , where  $g(x) = \frac{f(x)}{x+3}$  and  $f$  is a linear function. What is the  $y$ -intercept of the graph of  $y = f(x)$  in the  $xy$ -plane?

(A)  $(0, -9)$

(B)  $(0, 3)$

(C)  $(0, 9)$

(D)  $(0, 27)$

22



Mark for Review



For groups of 25 or more people, a museum charges \$28 per person for the first 25 people and \$17 for each additional person. Which function  $f$  gives the total charge, in dollars, for a tour group with  $n$  people, where  $n \geq 25$ ?

(A)  $f(n) = 17n + 275$

(B)  $f(n) = 17n + 28$

(C)  $f(n) = 17n + 700$

(D)  $f(n) = 45n - 425$